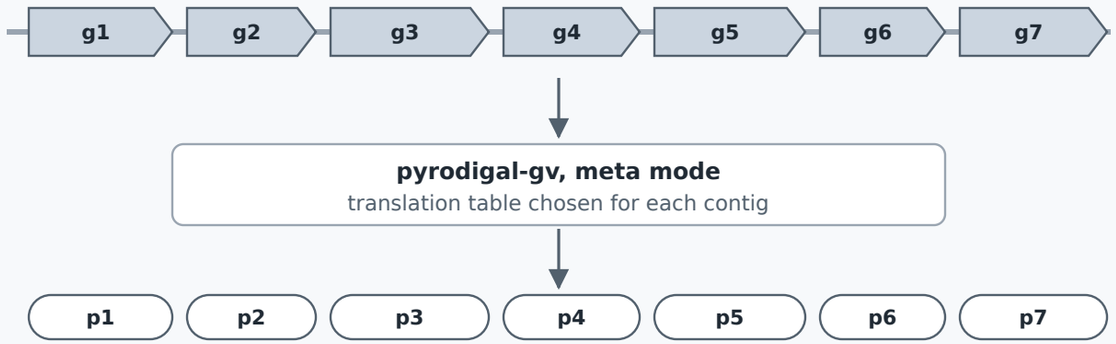
phu avger: finding auxiliary viral gene (AVG) candidates in viral contigs

A Gene prediction

Trusted viral contig (FASTA)



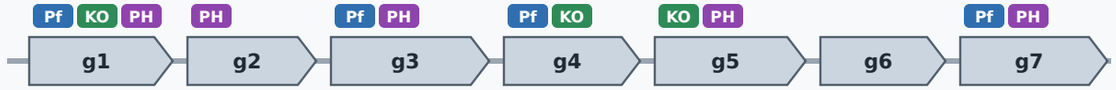
Translated proteins (at least 30 aa by default)

Predictions are cached and shared with phu screen and phu jack

B Two HMM annotation tracks (PyHMMER)

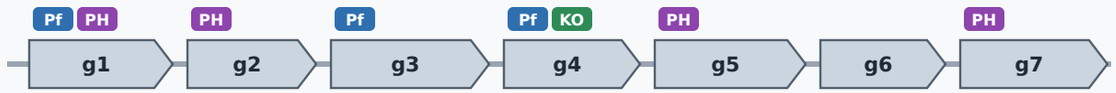
Pfam · KOfam · PHROG

Relaxed track E-value ≤ 1e-5; best hit per protein in each database



→ feeds V / VL scoring (panel C)

Strict track Pfam GA · KOfam thresholds · PHROG ≥ 60 bits · all: ≥ 0.5 of model and protein



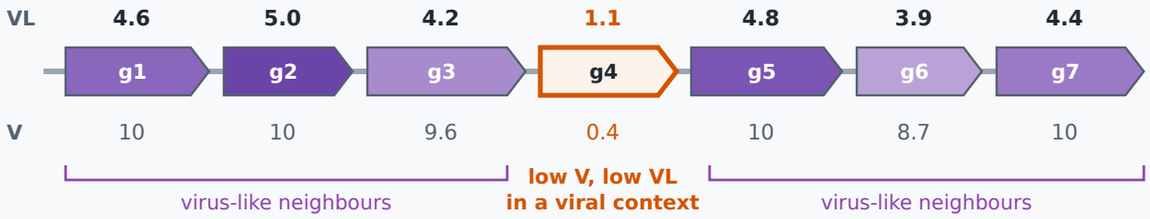
→ feeds functional calls and reference matching (panel D)

Each database is searched once; the two tracks differ only in the thresholds applied to the hits

Pfam, KOfam and PHROG are scored and averaged independently; values are never mixed

C Viral-likeness scores (V-Score-Search tables)

Relaxed-track hits are looked up in the V / VL table, separately for each database



contig AVL (per database) = mean VL of the contig's scored genes

here: (4.6 + 5.0 + 4.2 + 1.1 + 4.8 + 3.9 + 4.4) / 7 = 4.0

V-score how exclusively a protein family occurs in viruses (10 = virus only)

VL-score V-score weighted by how often the family is found in viruses

AVL average VL across a contig, per database: the viral context of a gene

VL shading: lower … higher below the gene cutoff

D Candidate test and curation

Candidate test, per database

contig AVL	**> 3**	Pfam, KOfam and PHRROG are each tested on their own
gene VL	**< 3**	
gene V	**< 10**	

fails in all → **not_candidate**

passes in at least one database → **AVG candidate**

1. **Match strict hits to the CheckAMG positive lists**
   AMG: Pfam, KOfam · APG and AReG: Pfam, KOfam, PHROG
2. **AMG weight ≥ 0.6 (--min-amg-weight)**
   applies to AMG evidence only
3. **Class filters (--filter-mode standard | strict | none)**
   an essential or structural hit in any database, PHROG included, removes the class

Final status

classified: putative_amg · putative_apg · putative_areg

class_conflict · filtered · below_amg_weight
unclassified_candidate · not_candidate

Outputs

avg-predictions.tsv	contig-summary.tsv	avg-proteins.faa	output-info.txt	detailed/ + .phu/run.json
one row per predicted AVG	genes, AVL and calls per contig	sequences of the predictions	counts and a data dictionary	per-gene tables and provenance

All calls are computational candidates for prioritisation, not confirmed auxiliary genes.

Scores: Zhou et al. 2026 · reference tables: CheckAMG · terminology: Martin et al. 2025 · gene values shown are illustrative